ABB ACCELERATOR 2026 · THEME 2

MULTIMODAL MAINTENANCE INTELLIGENCE AGENT

Torque Diagnostics

A citation-backed AI assistant for industrial troubleshooting, with NLP confidence on every answer.

NLP retrieval | Evidence-first answers | Calibrated confidence

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PROBLEM STATEMENT

Technicians should not hunt a 400-page manual while a machine is down.

Industrial maintenance teams depend on manuals, drawings, fault tables, wiring diagrams, and procedures. Traditional search cannot read that mix, so people guess, escalate, or wait on a senior technician.

Mixed documents

Text, scanned PDFs, tables, and diagrams live in different formats. Keyword search misses the page that actually matters.

Uncited answers

Generic AI can sound confident and still be wrong. On the plant floor, an uncited answer is an operational risk.

Lost expertise

Know-how sits in binders and in people’s heads. When senior technicians rotate, the plant loses troubleshooting speed.

Cost of delay

Every extra minute of downtime, unsafe bypass, or wrong spare part is expensive. The first answer has to be checkable.

PROPOSED SOLUTION

An AI chatbot that answers from manuals, diagrams, and images — then shows its work.

Torque Diagnostics is a technician-facing chatbot for Theme 2. Ask in text or send a photo of a nameplate, fault screen, or diagram. It retrieves evidence from those sources, writes a short answer, cites the page or figure, and scores confidence with NLP.

1. Ask

Type a fault, symptom, or “what do I check first?” — or drop a photo of a nameplate, HMI, or diagram into the chat.

2. Ground

Hybrid NLP search over manuals, tables, diagrams, and OCR’d scans or photos. Entities such as asset, fault, and procedure are extracted.

3. Answer with proof

Cited steps, source highlighting, and a High / Medium / Low NLP confidence score on every answer.

Theme 2 fit: grounded AI chatbot · PDF / image upload · OCR for scans and diagrams · text or photo questions · citation-backed · NLP confidence.

HOW A QUESTION IS ANSWERED

After sources are in the system, a text or image question follows this NLP path.

01
Retrieve
Hybrid search over the index and knowledge graph. A typed question or OCR text from a chat photo both hit.

›

02
Extract
NER pulls equipment, fault codes, parts, and procedure names from the question and the retrieved pages.

›

03
Answer
Write a short technician response using only retrieved passages. No unsupported steps.

›

04
Score
Calibrate confidence from retrieval strength, NLI entailment, and entity coverage.

›

05
Show
Chat answer, citations, highlighted source, and High / Medium / Low confidence.

Ingest, OCR, and graph build are the system path on the architecture slide. This slide is only what happens when a technician asks in chat — text or a photo.

DIFFERENTIATOR

Confidence is an NLP measurement, not a decoration.

Retrieval strength

How well the question matches the retrieved passages (semantic similarity + keyword overlap).

Answer–passage NLI

A Hugging Face entailment model checks whether the generated answer is actually supported by the cited text.

Entity coverage

Fault code, asset, and procedure mentioned in the question must appear in the evidence. Missing entities lower the score.

SCORE	WHAT THE TECHNICIAN SEES	BEHAVIOR
High	Cited procedure matches the fault and assets in the question.	Show steps + source highlight.
Medium	Partial match or one weak document.	Answer with caveats and extra sources.
Low	Weak retrieval or NLI contradiction.	Do not guess. Return passages and say evidence is insufficient.

ON THE FLOOR

Example: a drive fault during a shift.

Technician question

"ACS880 shows fault 2310 Overcurrent. What do I check first before replacing the drive?"

- Hybrid search hits the firmware fault-tracing entry for 2310.
- NER locks onto **ACS880**, **2310**, and **overcurrent**.
- Answer is generated only from those passages.

Torque Diagnostics response

Confidence: High

Check motor load and jamming, acceleration times, motor/cable condition, and Group 99 motor data before swapping the drive.

- Cite: ACS880 firmware manual, Fault tracing — 2310 Overcurrent
- Evidence viewer highlights the cause and "what to do" rows

WHY THIS THEME IS FAMILIAR

Current GWU coursework and industry work are how I understand ABB's Theme 2 problem.

I am not learning this problem only from the brief. The same mix of documents, diagnostics, and must-be-right answers already shows up in my courses and in systems I have shipped.

Academic coursework · M.S. Data Science, GWU

- **NLP:** manuals are language. Retrieval, NER on fault codes / assets, and NLI confidence are how a technician answer stays grounded.
- **Cloud computing:** plant document sets live in object storage and APIs, not on a laptop. Ingest has to be reliable.
- **Big data / warehousing:** a drive family has thousands of pages. Chunking, cataloging, and hybrid search are a data problem first.
- **ML / Deep Learning:** embeddings, ranking, and calibrated scores decide which page the technician sees first.

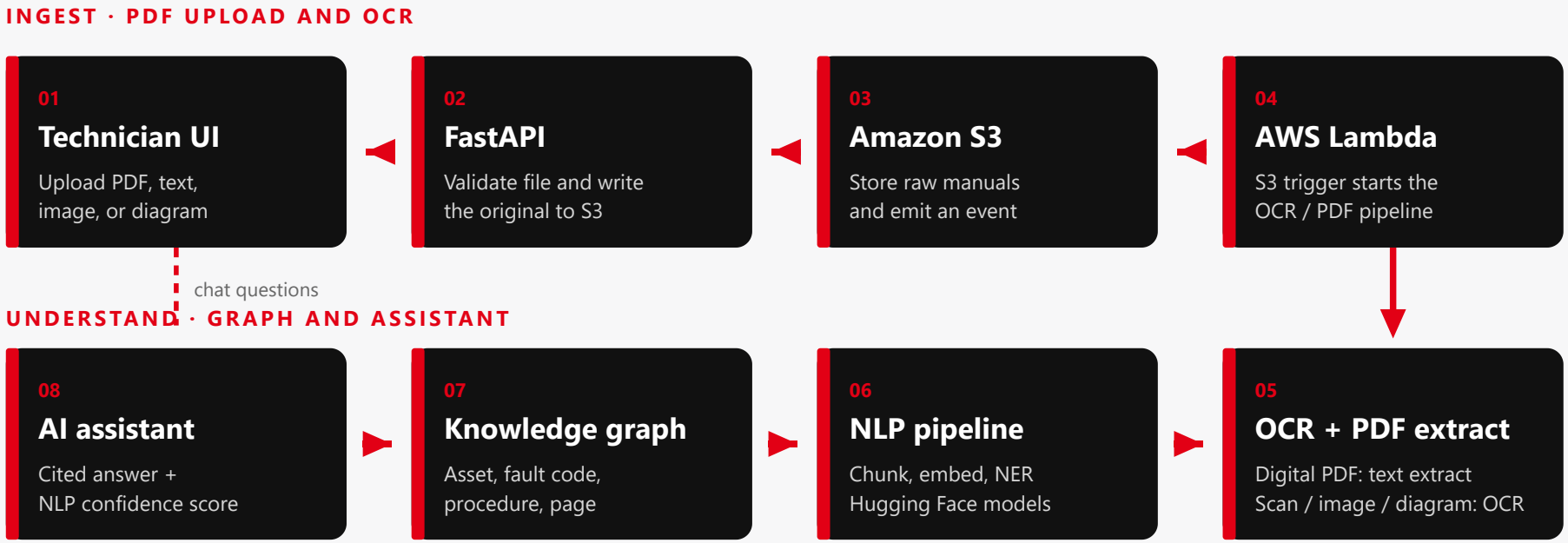
Career experience · 9+ years in industrial systems

- **FCA / Porsche / Bosch:** vehicle diagnostics, HMI, telemetry, and automotive operations — same world as motors, drives, and floor technicians.
- **Roche:** PDF / Excel diagnostic reporting. People make decisions from documents; the pipeline has to be trustworthy.
- **ETS:** high-volume, must-be-right event processing. A wrong answer is not acceptable in production.
- **IFF:** manufacturing and field / offline workflows. The assistant has to work the way a plant actually works.

ABB THEME 2 NEED	HOW I ALREADY UNDERSTAND IT
Citation-backed troubleshooting from manuals and tables	NLP + Roche document workflows: retrieve the page, show the evidence, do not guess.
Technician-facing answers under downtime pressure	FCA diagnostic UIs and Porsche/Bosch automotive operations: short, checkable first actions.
PDF + OCR ingest of mixed industrial documents	Theme 2 OCR requirement + Roche document pipelines + AWS S3/Lambda at ETS and Porsche.

ARCHITECTURE DIAGRAM

Torque Diagnostics system architecture



How files move

Digital PDF and text skip pixel OCR and use text extract. Scanned PDF, photos, and diagrams go through the OCR pipeline, then the same NLP and graph update.

The assistant reads the index plus the knowledge graph. Low NLP confidence means insufficient evidence, not a guessed procedure.

BUSINESS IMPACT AND PROTOTYPE PLAN

Checkable answers, shorter downtime, knowledge that stays on site.

Faster first action

Technicians get the relevant procedure and fault table without paging through binders.

Safer AI use

Cited answers reduce hallucinated maintenance steps on the floor.

Reusable plant knowledge

The same assistant can sit on ABB motors, drives, and automation document sets.

PROTOTYPE WINDOW	WHAT SHIPS FOR EVALUATION
Days 1–3	UI PDF / image upload → FastAPI → S3 → Lambda. Digital PDF extract + OCR on at least one scanned manual.
Days 4–7	NLP chunks, knowledge-graph update, FastAPI chat, Hugging Face NLI confidence, citations.
Days 8–10	Fault-code demo, demo video, README, and grounded vs. ungrounded evaluation questions.

FUTURE SCOPE · AFTER THE PROTOTYPE

Grow only where the floor needs it.

Richer tables

Deeper fault-table structure as more manuals are uploaded.

Diagram vision

Wiring-drawing understanding beyond OCR text on the page.

Field mode

Offline / low-connectivity assistance using prior Electron / SQLite work.

Larger corpus

Batch indexing with PySpark if document volume leaves a single-node job.

READY TO PROTOTYPE

Torque Diagnostics

Ask from the source. Cite the page. Show the confidence.

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